

The Loyola Index Family and the Resolution Limit of Degree Imbalance

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Abstract

The two-parameter Gutman–Milovanović family $M_{\alpha,\beta}(G) = \sum_{uv \in E(G)} (d_u d_v)^\alpha (d_u + d_v)^\beta$ unifies the Zagreb, Randić, sum-connectivity and related degree-based indices. We study its normalised degree-imbalance exponential extension, the *Loyola index family*

$$LO(G; \alpha, \beta, \gamma) = \sum_{uv \in E(G)} (d_u d_v)^\alpha (d_u + d_v)^\beta \exp\left(\gamma \frac{|d_u - d_v|}{d_u + d_v}\right),$$

with $LO(G; \alpha, \beta, 0) = M_{\alpha,\beta}(G)$ identically. We ask what the imbalance coordinate γ adds to the parent family. On a ratio plane that separates size from shape, the pure- γ points distinguish two graphs exactly when their histograms of edge imbalances differ, a limit shared by every index that depends on an edge only through its degree ratio. On small trees this is the discrimination limit of all degree-based indices, and we prove that it is lost from order thirteen. We bound the number of times two graphs exchange order along γ , characterise the directions in parameter space that a given graph can identify, and determine the extremal trees of the pure- γ points. A gain from γ on vaporisation enthalpies that replicated

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under a pre-registered protocol disappears once the parent family is searched over a box that does not truncate it. Pre-registered tests on measured boiling points show no demonstrated predictive gain from γ over the parent family.

1 Introduction

Degree-based topological indices form one of the most developed descriptor classes of chemical graph theory. The class runs from the Randić and Zagreb indices [18, 32] and their parametric extensions, the general Randić and sum-connectivity families [6, 41], to the Sombor index [14] and its many variants, surveyed in [15]. Extremal results on bond-incident-degree (BID) indices are surveyed in [2], while a finite-dimensional linear framework for such indices is developed in [30].

The *Gutman–Milovanović family*, introduced in [17] and studied in the recent bounds and QSPR literature [13, 24, 28], unifies the product-based and sum-based degree indices with two parameters. It is defined by $M_{\alpha,\beta}(G) = \sum_{uv \in E(G)} (d_u d_v)^\alpha (d_u + d_v)^\beta$, $\alpha, \beta \in \mathbb{R}$. Its special cases are catalogued in [24]. They include the first and second Zagreb indices $M_{0,1} = M_1$ and $M_{1,0} = M_2$, the hyper-Zagreb index $M_{0,2} = HM$ [35], the modified second Zagreb index $M_{-1,0} = {}^mM_2$ and the Randić index $M_{-1/2,0} = R$. Further special cases are the sum-connectivity index $M_{0,-1/2} = \chi$ [40], the harmonic index $2M_{0,-1} = H$, the inverse sum indeg index $M_{1,-1} = ISI$, the geometric–arithmetic index $2M_{1/2,-1} = GA$ [37] and the arithmetic–geometric index $\frac{1}{2}M_{-1/2,1} = AG$. The general Randić family $R_\alpha = M_{\alpha,0}$ and the general sum-connectivity family $\chi_\beta = M_{0,\beta}$ are its two parameter axes, and parameter-optimised versions of the family have already been fitted to octane properties [3]. The family, its special cases and its tunability are therefore not new.

In this paper we study a three-parameter extension of $M_{\alpha,\beta}$, the *Loyola index family*, obtained by attaching a separately parameterised exponential factor in the normalised degree imbalance $q_{uv} = |d_u - d_v|/(d_u + d_v)$ of each edge. The definition is $LO(G; \alpha, \beta, \gamma) = \sum_{uv} (d_u d_v)^\alpha (d_u + d_v)^\beta e^{\gamma q_{uv}}$, so that $LO(G; \alpha, \beta, 0) = M_{\alpha,\beta}(G)$ identically, and positive γ exponentially upweights degree-imbalanced edges while negative γ downweights them.

On regular graphs γ has no effect. The paper therefore concerns γ alone. We ask whether the imbalance direction is independent of the parent parameters, what analytic structure it carries, and whether it adds structural or predictive information beyond an optimised Gutman–Milovanović parent.

Writing the summand as $\exp(\alpha \ln(d_u d_v) + \beta \ln(d_u + d_v) + \gamma q_{uv})$ shows that the family is a three-parameter subfamily of the exponential vertex-degree-based indices of Rada [29], whose extremal theory is developed in [8, 12, 19, 36]. Beyond the extremal trees of the pure- γ points (Section 5.2), extremal problems are not considered here. Published one-parameter exponential indices (e.g. [9, 10]) place their parameter on a single classical generator inside the exponent. Here the two Gutman–Milovanović basis terms act as the factors $(d_u d_v)^\alpha (d_u + d_v)^\beta$, while the imbalance coordinate carries its own parameter γ . The imbalance ratio itself is not new. It appears in the diminished Sombor index [25] and, as a normalised form of Albertson’s irregularity [1], in the irregularity index $IRLA(G) = 2 \sum_{uv} q_{uv}$ [33]. The pure- γ point $LO(G; 0, 0, \gamma) = \sum_{uv} e^{\gamma q_{uv}}$ is the exponential index, in the sense of [29], of the edge function γq_{uv} , and at $\gamma = 2$ it is the exponential of $IRLA$. The ratio also appears in degree-ratio indices [16, 27] and in the parent family itself, since $1 - q_{uv}^2 = 4d_u d_v / (d_u + d_v)^2$ gives $M_{\alpha, -2\alpha}(G) = 4^{-\alpha} \sum_{uv} (1 - q_{uv}^2)^\alpha$. What is new is a coordinate *linear* in q_{uv} in the exponent. Unlike $\ln(1 - q_{uv}^2)$, this coordinate is not an affine function of $\ln(d_u d_v)$ and $\ln(d_u + d_v)$ once a vertex of degree at least 3 is admitted. We prove this below. To our knowledge, no published index couples it multiplicatively, with its own parameter, to the Gutman–Milovanović family. The exponential form is not derived from a physical model and is chosen instead because it makes the family log-linear in its parameters. Then γ acts multiplicatively and only on irregularity, and the whole family is an exponential family in (α, β, γ) , which makes available the convexity, identifiability and discrimination results that follow.

On hydrogen-suppressed molecular graphs with $\Delta \leq 4$, every BID index is a linear functional of the ten edge-degree-pair counts $m_{ij}(G) = |\{uv \in E(G) : \{d_u, d_v\} = \{i, j\}\}|$, $1 \leq i \leq j \leq 4$ (compare the linear

framework of Rada [30]). Hence the Loyola family lies inside a fixed ten-dimensional descriptor space, however its parameters are chosen.

The tools are classical (log-sum-exp convexity, Laguerre’s rule of signs, the Lindemann–Weierstrass theorem, exponential-family minimality); the contribution is the exact characterisation they yield, which also bounds every index that sees an edge only through its degree ratio. The family is not proposed as a better predictor, and we find no demonstrated predictive advantage on the sets and protocols studied here. Its interest lies in the structure it exposes. It keeps the interpretation of the parent family, separates size from shape, and shares the ratio-plane geometry of GA and AG . Because the QSPR datasets are small (18 octanes, 35 nonanes, 34 decanes, 100 pooled C6–C10 alkanes), the chemometric results are diagnostic evidence rather than proof of chemical predictive utility.

The contributions are as follows. (i) The scale–shape factorisation and the ratio-plane theorem (Theorem 2), which characterises exactly what the pure- γ line can discriminate, with tree collisions at every order from 13, and from 16 for $\Delta \leq 4$. Its corollary extends the collisions to every index that depends on an edge only through its degree ratio, such as GA , AG and the diminished Sombor index. (ii) The crossing bound (Proposition 3, from Laguerre’s rule of signs) and per-graph identifiability (Proposition 4), together with the independence lemma for the admissible pair set (Lemma 1). (iii) Log-convexity with mean/covariance derivatives (Theorem 1), closed forms on standard graphs, termwise bounds for $\gamma \geq 0$ and the extremal trees of the pure- γ points (Section 5). (iv) The first orders at which discrimination is lost on trees, with the colliding profiles, the degree-pair ceiling realised on the octane benchmark (Proposition 6), and structure sensitivity. (v) A reproducible QSPR diagnostic, in which freeing γ gives no demonstrated gain once the parent’s search box does not truncate it, complemented by pre-registered tests on measured boiling points of decanes and of pooled C6–C10 alkanes (Section 6.4). There, ridge regression on the degree-pair counts is far ahead of every single index on the decanes and ahead on the pooled set.

2 Preliminaries

2.1 Notation

Let $G = (V, E)$ be a finite simple connected graph of order $n = |V|$ and size $m = |E|$, and for a vertex $v \in V$ let the degree $d_v = d_G(v)$ count the edges incident to v . The maximum and minimum degrees are denoted $\Delta = \Delta(G)$ and $\delta = \delta(G)$. The chemometric sections use hydrogen-suppressed molecular graphs of alkane carbon skeletons, for which $\Delta \leq 4$, while the mathematical results do not assume this bound unless stated.

For an edge $e = uv \in E(G)$ we write (d_u, d_v) for its *degree pair*, understood as an unordered multiset $\{d_u, d_v\}$.

2.2 Definition

Definition 1 (Loyola index family). Let $G = (V, E)$ be a simple connected graph. For $\alpha, \beta, \gamma \in \mathbb{R}$, the *Loyola index family* is defined as

$$LO(G; \alpha, \beta, \gamma) = \sum_{uv \in E(G)} (d_u d_v)^\alpha (d_u + d_v)^\beta \exp\left(\gamma \frac{|d_u - d_v|}{d_u + d_v}\right) \quad (1)$$

Eq. (1) is well defined for all $\alpha, \beta \in \mathbb{R}$ on graphs without isolated vertices. Each summand is symmetric in (u, v) , so $LO(G; \alpha, \beta, \gamma)$ depends only on the multiset of unordered edge degree pairs of G . When $\gamma = 0$ the family reduces identically to the Gutman–Milovanović family [17, 24],

$$LO(G; \alpha, \beta, 0) = M_{\alpha, \beta}(G) = \sum_{uv \in E(G)} (d_u d_v)^\alpha (d_u + d_v)^\beta,$$

whose classical special cases are catalogued in the Introduction. Throughout, $q_{uv} = \text{imb}(d_u, d_v)$, where $\text{imb}(x, y) = |x - y|/(x + y)$, denotes the normalised degree imbalance of the edge uv , and positive γ exponentially upweights degree-imbalanced edges while negative γ downweights them. On regular graphs every $q_{uv} = 0$, so γ has no effect and $LO(G; \alpha, \beta, \gamma) = M_{\alpha, \beta}(G)$ for all γ .

3 Parameter geometry

Throughout, $z_{uv} = (\ln(d_u d_v), \ln(d_u + d_v), q_{uv})$ is the feature vector of the edge uv and $\theta = (\alpha, \beta, \gamma)$, so that $LO_G(\theta) = \sum_{uv \in E(G)} e^{\theta^\top z_{uv}}$.

Lemma 1 (Affine independence of the imbalance direction). *For $\Delta \geq 3$, with $P_\Delta = \{(i, j) : 1 \leq i \leq j \leq \Delta\}$ the admissible degree pairs, the augmented feature vectors $(1, \ln(ij), \ln(i + j), q_{ij})$, $(i, j) \in P_\Delta$, span $\mathbb{R}^4$. Consequently, if two parameter triples θ, θ' satisfy $e^{\theta^\top z} = c e^{\theta'^\top z}$ for some constant $c > 0$ and all $(i, j) \in P_\Delta$, then $\theta = \theta'$ and $c = 1$. In particular, no choice of (α', β') and scaling constant reproduces the action of $\gamma \neq 0$ on all admissible degree pairs. For $\Delta = 2$ the span has dimension 3, while for $\Delta = 1$ the only pair is $(1, 1)$, with augmented vector $(1, 0, \ln 2, 0)$, and the span has dimension 1. In both cases the conclusion fails; for $\Delta = 2$, indeed, $q_{ij} = [\ln(i + j) - \ln 2 - \frac{1}{2} \ln(ij)] / (3 \ln 3 - \frac{9}{2} \ln 2)$ on the three pairs, so there the γ direction is a rescaled (α, β) direction.*

Proof. The vectors $(1, \ln(ij), \ln(i + j))$ on the pairs $(1, 1), (2, 2), (1, 2)$ have rank 3, since the corresponding 3×3 determinant equals $\ln 2 \cdot \ln(9/8) \neq 0$, so it suffices to show that q_{ij} is not an affine function of $(\ln(ij), \ln(i + j))$ on P_Δ . The four pairs $(1, 1), (2, 2), (1, 2), (1, 3)$, all admissible once $\Delta \geq 3$, witness this. Suppose $q_{ij} = a + b \ln(ij) + c \ln(i + j)$ on these pairs. The pairs $(1, 1)$ and $(2, 2)$ have $q = 0$, so they give $a + c \ln 2 = 0$ and $a + (2b + 2c) \ln 2 = 0$, hence $a = -c \ln 2$ and $b = -c/2$. Substituting into the $(1, 2)$ and $(1, 3)$ equations gives $\frac{1}{3} = c(\ln 3 - \frac{3}{2} \ln 2)$ and $\frac{1}{2} = c(\ln 2 - \frac{1}{2} \ln 3)$. In particular $c \neq 0$. Three times the first minus twice the second gives $0 = c(4 \ln 3 - \frac{13}{2} \ln 2)$, hence $8 \ln 3 = 13 \ln 2$, i.e. $3^8 = 2^{13}$. This is false ($6561 \neq 8192$). Hence the augmented vectors have rank 4. If $e^{\theta^\top z} = c e^{\theta'^\top z}$ on P_Δ , then $(\theta - \theta')^\top z_{ij} = \ln c$ for all admissible pairs, so $(\ln c, \theta' - \theta)$ is orthogonal to every augmented vector, and $\theta = \theta'$ and $c = 1$. For $\Delta = 2$ only the pairs $(1, 1), (1, 2), (2, 2)$ exist, and a direct computation shows the augmented vectors have rank 3, while for $\Delta = 1$ the single augmented vector has rank 1. ■

Remark (Independence versus per-graph identifiability). Lemma 1 concerns the edge-weight *function* on the full admissible pair set rather than

every individual graph. On a single graph G some direction v satisfies $LO(G; \theta + tv) = e^{ct} LO(G; \theta)$ for all t , unless the realised feature vectors z_{ij} affinely span $\mathbb{R}^3$ (Proposition 4). Along such a direction the index only rescales. On a regular graph nothing about θ is identifiable beyond one linear functional. If the edges of G realise at most three affinely independent augmented vectors, some direction of θ , possibly the γ direction, is unidentifiable from $LO(G; \cdot)$ alone. The lemma guarantees that no such degeneracy is forced on the whole molecular class $\Delta \geq 3$, so the family separates parameters as a family of descriptors, although a particular graph may not.

The next result is the standard convexity of a log-sum-exp (log-partition) function, stated for LO because strictness depends on the degree pairs that G realises.

Theorem 1 (Global log-convexity). *Fix a graph G with $m \geq 1$ edges, define $F_G(\theta) = \log LO_G(\theta)$ on $\mathbb{R}^3$, and let $\mathbb{E}_\theta$ and Cov_θ denote mean and covariance under the edge probability measure $w_{uv}(\theta) = e^{\theta^\top z_{uv}} / LO_G(\theta)$. Then F_G is convex, with*

$$\nabla F_G(\theta) = \mathbb{E}_\theta[z], \quad \nabla^2 F_G(\theta) = \text{Cov}_\theta(z) \succeq 0.$$

In particular $\partial_\gamma F_G = \mathbb{E}_\theta[q] \in [0, 1)$ and $\partial_\gamma^2 F_G = \text{Var}_\theta(q) \geq 0$, so $\log LO$ is non-decreasing and convex in γ , and it is strictly increasing in γ unless every edge of G joins vertices of equal degree. Moreover $\text{Var}_\theta(q) = 0$ at some (equivalently, every) θ if and only if q_{uv} is constant over the edges of G , which covers regular graphs ($q \equiv 0$) and, for example, (Δ, δ) -semiregular bipartite graphs. On these every edge carries the same nonzero imbalance, and $\log LO$ is affine and strictly increasing in γ . F_G is strictly convex if and only if the edge feature vectors $\{z_{uv}\}_{uv \in E}$ do not lie in a proper affine subspace of $\mathbb{R}^3$. On a regular graph $F_G(\theta) = \log m + \theta^\top z_0$ is affine and $\text{Cov}_\theta(z) = 0$.

Proof. Direct differentiation of $LO_G(\theta) = \sum_{uv} e^{\theta^\top z_{uv}}$ gives $\partial_k F_G = \sum_{uv} w_{uv} z_{uv,k} = \mathbb{E}_\theta[z_k]$ and $\partial_k \partial_l F_G = \mathbb{E}_\theta[z_k z_l] - \mathbb{E}_\theta[z_k] \mathbb{E}_\theta[z_l]$, the (k, l) -entry of $\text{Cov}_\theta(z)$, which is positive semidefinite. Convexity follows. The measure $w(\theta)$ has full support on E for every θ . Hence $\text{Cov}_\theta(z)$ is

singular in a direction v iff $v^\top z_{uv}$ is constant over all edges, i.e. iff the z_{uv} lie in a proper affine subspace, which proves the strictness criterion. Since $q_{uv} \in [0, 1)$, $\mathbb{E}_\theta[q] \in [0, 1)$, and $\mathbb{E}_\theta[q] = 0$ iff $q_{uv} = 0$ on every edge. On an r -regular graph all z_{uv} equal $z_0 = (2 \ln r, \ln 2r, 0)$. ■

The Cauchy–Schwarz midpoint inequality of Section 5 is midpoint log-convexity, a corollary of this theorem.

3.1 The ratio plane, crossings and identifiability

Since $4ij/(i+j)^2 = 1 - q_{ij}^2$, every admissible pair satisfies $\ln(ij) = 2 \ln(i+j) + \ln(1 - q_{ij}^2) - \ln 4$, and therefore

$$LO(G; \alpha, \beta, \gamma) = 4^{-\alpha} \sum_{uv \in E(G)} (d_u + d_v)^{2\alpha+\beta} (1 - q_{uv}^2)^\alpha e^{\gamma q_{uv}}. \quad (2)$$

Thus, in the coordinates (κ, α, γ) with $\kappa = 2\alpha + \beta$, the parameter κ acts only on the edge degree sum (scale), while α and γ act only on q_{uv} , a function of the degree ratio (shape). Write $n_q(G)$ for the number of edges of G with $q_{uv} = q$ (the q -histogram) and $\Pi = \{(\alpha, \beta, \gamma) : \beta = -2\alpha\}$ for the plane $\kappa = 0$, which contains every pure- γ point $(0, 0, \gamma)$, $GA = 2 LO(\frac{1}{2}, -1, 0)$ and $AG = \frac{1}{2} LO(-\frac{1}{2}, 1, 0)$.

Theorem 2 (Ratio plane). *Let G and H be graphs.*

- (i) *On Π , $LO(G; \alpha, -2\alpha, \gamma) = \sum_q n_q(G) ((1 - q^2)/4)^\alpha e^{\gamma q}$ depends on G only through its q -histogram, and G and H agree at every point of Π if and only if $n_q(G) = n_q(H)$ for all q . Otherwise their values differ at every point of Π outside a set of measure zero.*
- (ii) *For every nonzero algebraic γ , $LO(G; 0, 0, \gamma) = LO(H; 0, 0, \gamma)$ if and only if $n_q(G) = n_q(H)$ for all q . On any finite set of graphs the same holds for all but finitely many real γ .*
- (iii) *Let G and H be trees with distinct edge-degree-pair profiles and equal q -histograms, each having a leaf adjacent to a vertex of the same degree $d \geq 2$. Attaching a pendant path of length $L \geq 1$ at such a leaf in each produces trees with the same two properties (the profiles*

change by the same vector, whether or not the chosen leaf edge was the only $(1, d)$ -edge) and the same maximum degree. Consequently, for every $n \geq 13$ (and every $n \geq 16$ among trees with $\Delta \leq 4$) there are non-isomorphic trees of order n with distinct degree-pair profiles that no point of Π separates.

Proof. (i) is immediate from (2), since setting $\beta = -2\alpha$ removes the degree-sum factor. For (ii) and (iii), agreement for equal histograms is immediate. Suppose the histograms differ. At $\alpha = 0$ the difference is $\sum_q (n_q(G) - n_q(H)) e^{\gamma q}$, an exponential sum with distinct exponents and not all coefficients zero, so it is not identically zero. The difference on Π is therefore a nonzero real-analytic function of (α, γ) , whose zero set has measure zero. (ii) For algebraic $\gamma \neq 0$ the exponents γq , $q \in \mathbb{Q}$, are distinct algebraic numbers, so by the Lindemann–Weierstrass theorem [4] the numbers $e^{\gamma q}$ are linearly independent over the algebraic numbers. Hence the integer coefficients $n_q(G) - n_q(H)$ must all vanish. For real γ , a nonzero exponential sum with k distinct real exponents has at most $k - 1$ real zeros (Proposition 3 below), and a finite set of graphs gives finitely many such sums. (iii) The attachment changes the profile of each tree by the same vector, $-(1, d) + (2, d) + (L - 1)(2, 2) + (1, 2)$, so distinct profiles stay distinct and equal histograms stay equal. The new or changed vertices have degree at most 2, so the maximum degree is unchanged when it is at least 2. Trees realising the two order-13 profiles of Section 4.2 are the following. T_1 : a path $c a b d e \ell$ in which c carries five pendant leaves, b and d carry one pendant leaf each and ℓ is a leaf, so that $d_c = 6$, $d_a = d_e = 2$, $d_b = d_d = 3$; its edges are $(1, 6)^5, (2, 6), (2, 3)^2, (3, 3), (1, 3)^2, (1, 2)$. T_2 : a path $c b x y z$ in which c carries five pendant leaves, b one and z two, so that $d_c = 6$, $d_b = d_z = 3$, $d_x = d_y = 2$; its edges are $(1, 6)^5, (3, 6), (1, 3)^3, (2, 3)^2, (2, 2)$. Both have a leaf adjacent to a vertex of degree $d = 3$. Consider now the order-16 pair with $\Delta \leq 4$. The profile $\{(1, 2)^2, (1, 4)^8, (2, 4)^2, (4, 4)^3\}$ is realised by a path $A B C D$ of vertices of degree 4. Here A and D carry three pendant leaves each, while B and C carry one pendant leaf and one pendant path of length 2 each. The profile $\{(1, 4)^8, (2, 2)^3, (2, 4)^4\}$ is realised by three vertices A, B, C of degree 4. They are joined as $A w B$ through a vertex w of degree 2 and as $B v_1 v_2 v_3 v_4 C$ through a path of four vertices

of degree 2, and A , B and C carry three, two and three pendant leaves. Both have a leaf adjacent to a vertex of degree $d = 4$, which gives every larger order. ■

The first sentence of part (ii) is the Lindemann–Weierstrass argument of Rada [29] applied to the edge function γ_{uv} , whose equivalence classes of graphs are exactly the graphs with equal q -histograms; it is included for completeness. Parts (i) and (iii), the statement for real γ and the corollary below are new.

Corollary (Degree-ratio indices). *Let $\rho_{uv} = \min(d_u, d_v) / \max(d_u, d_v)$ and let $I_f(G) = \sum_{uv \in E(G)} f(\rho_{uv})$ for any function f on $(0, 1]$. Since $\rho_{uv} = (1 - q_{uv}) / (1 + q_{uv})$, I_f depends on G only through its q -histogram. Consequently the trees of Theorem 2(iii), which exist at every order $n \geq 13$ (and $n \geq 16$ with $\Delta \leq 4$), are not separated by any such index. This applies to GA , AG , the diminished Sombor index [25], the symmetric division deg index [38], the inverse symmetric division deg index and its exponential [10], and the degree-ratio indices [16, 27].*

The corollary needs no proof beyond the identity for ρ_{uv} ; it turns the colliding families of part (iii) into a limitation for the whole class of indices that see an edge only through its degree ratio. Among the members of that class, the pure- γ points are those for which the limitation is also sharp, by part (ii).

Theorem 2 is both a limitation result and a characterisation. By exhaustive enumeration of orders 7–17 (orders below 7 are trivial), q -histograms coincide with degree-pair profiles for all trees of order at most 12. The same holds for trees with $\Delta \leq 4$ of order at most 15. So the pure- γ points attain the degree-pair floor exactly up to those orders and, by part (iii), at no larger order. The tie of GA , AG and the pure- γ points on the trees of order 10 (Section 4.2) is consistent with this, since all lie on Π . At orders 13–17 generic points off Π (Proposition 7) and the Randić index separate more trees than any point of Π . In the other direction, GA , whose values are sums of square roots, merges two distinct q -histograms among trees first at order 17. A witness is the pair $\{(1, 5)^{10}, (2, 2)^2, (5, 5)^2, (2, 5), (1, 2)\}$ and

$\{(4, 5)^3, (1, 4)^5, (1, 5)^6, (2, 5), (1, 2)\}$, which have equal GA but different q -histograms, while at that order the pure- γ points, by part (ii), still separate all histograms. That GA depends only on the degree ratio is well known, and this paper adds the exact ceiling and the colliding families.

Proposition 3 (γ -crossings). *Fix (α, β) and graphs G, H , let $q_1 < \dots < q_r$ be the distinct imbalances of the degree pairs occurring in G or H , and write $D(\gamma) = LO(G; \alpha, \beta, \gamma) - LO(H; \alpha, \beta, \gamma) = \sum_{k=1}^r c_k e^{\gamma q_k}$ with $c_k = \sum_{q_{ij}=q_k} (m_{ij}(G) - m_{ij}(H))(ij)^\alpha (i+j)^\beta$. Let V be the number of sign changes in $(c_1, \dots, c_r)$ after zero terms are removed. If $D \not\equiv 0$, then D has at most $V \leq r - 1$ real zeros counted with multiplicity, and their number has the parity of V . In particular, for graphs with $\Delta \leq 4$ the possible imbalances are $0, \frac{1}{7}, \frac{1}{5}, \frac{1}{3}, \frac{1}{2}, \frac{3}{5}$, and two such graphs exchange order at most five times as γ ranges over $\mathbb{R}$, for every (α, β) .*

Proof. With $x = e^\gamma > 0$, D becomes the generalised polynomial $\sum_k c_k x^{q_k}$ with distinct real exponents, so Laguerre's extension of Descartes' rule of signs applies [20]. The number of positive zeros, with multiplicity, is at most V and has its parity, and the parity statement also follows from the signs of D as $\gamma \rightarrow \mp\infty$. These are the signs of the first and last nonzero c_k , by the termwise argument of Proposition 5 below applied to D . ■

The bound V can be attained. Two trees of order 12 with maximum degree 6 and eight distinct imbalances, with profiles $\{(1, 3)^2, (1, 4)^2, (1, 6)^5, (3, 4), (4, 6)\}$ and $\{(1, 2), (1, 5)^4, (1, 6)^4, (2, 6), (5, 6)\}$, have $V = 5$ and cross five times at $(\alpha, \beta) = (-1, 0)$, near $\gamma = -17.2, -12.0, -2.0, 2.7$ and 31.1 . By the bound these are all its zeros. Whether five crossings are attained for $\Delta \leq 4$ is open. Along directions other than the γ axis the exponents are not the q_k , so the constant five does not apply.

The per-graph identifiability discussed in the remark after Lemma 1 can be made exact. Let $R(G)$ be the set of degree pairs realised by G , $r(G)$ the number of distinct imbalances among them, and $A(G)$ the affine dimension of $\{z_{ij} : (i, j) \in R(G)\}$.

Proposition 4 (Per-graph scale-invariant directions). *(i) $A(G)$ equals the rank of the Hessian $\text{Cov}_\theta(z)$ of $\log LO_G$ at every θ , and the directions v with $LO(G; \theta + tv) = e^{ct} LO(G; \theta)$ for all t and θ form a subspace of dimension $3 - A(G)$.*

(ii) $A(G) \leq \min(|R(G)| - 1, r(G), 3)$ for every graph.

(iii) If $r(G) = 2$ with imbalances $q_1 \neq q_2$, then $v = (1, -2, c)$ with $c = [\ln(1 - q_2^2) - \ln(1 - q_1^2)]/(q_1 - q_2)$ is such a direction.

Proof. (i) is the standard minimal-representation property of exponential families [7], and since $\text{Cov}_\theta(z)$ has full support on $R(G)$, its rank is the affine dimension of the support. Also $v^\top z$ is constant on $R(G)$ exactly for v orthogonal to the differences of the z_{ij} . (ii) By (2), $z_{ij} = (2s + \ln(1 - q^2) - \ln 4, s, q)$ with $s = \ln(i + j)$, so pairs of equal imbalance lie on one line parallel to $(2, 1, 0)$, and r parallel lines span an affine space of dimension at most r . (iii) $v^\top z_{ij} = \ln(1 - q_{ij}^2) - \ln 4 + c q_{ij}$ takes the same value at q_1 and q_2 by the choice of c . ■

Equality holds in (ii) for every set of admissible pairs with $\Delta \leq 6$ (by exhaustive enumeration), hence for all molecular graphs. It fails from $\Delta = 7$. The single violating set of at most four pairs there is $\{(1, 7), (2, 6), (3, 5), (4, 4)\}$, whose degree sums all equal 8. A constant degree sum makes $\ln(i + j)$ constant on the set, so $A \leq 2$, while the bound equals 3 as soon as the set has at least four pairs and three imbalances. A connected graph all of whose edges have the same degree sum s realises only one pair. Indeed, each neighbour of a vertex of degree d then has degree $s - d$, so by connectivity only the pair $\{d, s - d\}$ occurs. Hence no connected graph has this set as its pair set. As an illustration of (iii), the spider with four legs of length three realises the pairs $(2, 4)$, $(2, 2)$ and $(1, 2)$, which have only two imbalances ($\frac{1}{3}$ and 0). On it, γ is confounded with (α, β) along the direction of part (iii). For a QSPR dataset the relevant set is the union of the pairs realised by its molecules. A direction along which every molecule’s index is rescaled by the same factor is invisible to any correlation or regression on the dataset. This concerns identifiability of θ from the index values on the given set of graphs rather than identifiability

of a fitted regression model. On the octanes this union contains nine of the ten admissible pairs and its affine dimension is 3, so no such direction exists.

The behaviour at the two ends of the γ axis is used in the proof of Proposition 3. Routine termwise bounds in (m, Δ, δ) , a Cauchy–Schwarz interpolation bound and a uniform ratio bound are collected in Section 5.3.

Proposition 5 (Limits along the γ axis). *Let $E_{\max}(G)$ denote the set of edges realising $\text{imb}_{\max}(G) = \max_{uv} \text{imb}(d_u, d_v)$ and $E_{\min}(G)$ the set realising $\text{imb}_{\min}(G) = \min_{uv} \text{imb}(d_u, d_v)$. Then as $\gamma \rightarrow +\infty$,*

$$\frac{LO(G; \alpha, \beta, \gamma)}{e^{\gamma \text{imb}_{\max}(G)}} \longrightarrow \sum_{uv \in E_{\max}(G)} (d_u d_v)^\alpha (d_u + d_v)^\beta,$$

and as $\gamma \rightarrow -\infty$,

$$LO(G; \alpha, \beta, \gamma) / e^{\gamma \text{imb}_{\min}(G)} \longrightarrow \sum_{uv \in E_{\min}(G)} (d_u d_v)^\alpha (d_u + d_v)^\beta.$$

Proof. At $\gamma = 0$ the exponential factor is 1 on every edge. With $\mu(G) := \text{imb}_{\max}(G)$ and $\varepsilon_{uv} := \mu(G) - \text{imb}(d_u, d_v) \geq 0$, we have $LO(G; \alpha, \beta, \gamma) / e^{\gamma \mu(G)} = \sum_{uv} (d_u d_v)^\alpha (d_u + d_v)^\beta e^{-\gamma \varepsilon_{uv}}$. The summands over $E_{\max}(G)$ ($\varepsilon_{uv} = 0$) do not depend on γ , while the others ($\varepsilon_{uv} > 0$) tend to 0 as $\gamma \rightarrow +\infty$, termwise since $E(G)$ is finite. The case $\gamma \rightarrow -\infty$ is dual. ■

4 Discrimination

4.1 The degree-pair ceiling

No vertex-degree-based index separates graphs with equal degree-pair counts, so the number of distinct profiles is a *ceiling* on distinct values and a *floor* on degeneracy.

Proposition 6 (Degree-pair information ceiling). *If two graphs G and H have identical unordered edge-degree-pair count vectors, $m_{ij}(G) = m_{ij}(H)$*

for all $i \leq j$, then $LO(G; \theta) = LO(H; \theta)$ for every $\theta \in \mathbb{R}^3$, and more generally $TI(G) = TI(H)$ for every vertex-degree-based index TI . Among the 18 octane isomers exactly 16 distinct count vectors occur, and the pairs $\{3\text{-methylheptane}, 4\text{-methylheptane}\}$ and $\{3,4\text{-dimethylhexane}, 3\text{-ethyl-2-methylpentane}\}$ share their count vectors but differ in measured properties (Table 1). Hence every vertex-degree-based descriptor, tuned or not, carries a nonzero irreducible error on this benchmark.

Proof. By Definition 1, $LO(G; \theta)$ depends on G only through the multiset of unordered edge degree pairs, that is, through $(m_{ij}(G))_{i \leq j}$, and the same holds for every vertex-degree-based index [30]. The counts for the 18 octane isomers are computed and cross-checked in the reproducibility package, and Table 1 lists the common count vectors of both collision pairs. ■

The ceiling is tight. Distinct count vectors are separated at almost every parameter choice.

Proposition 7 (Generic discrimination of distinct profiles). *Fix $\Delta \geq 1$ and let $m \neq m'$ be two distinct edge-degree-pair count vectors supported on the admissible pairs P_Δ . Then $D(\theta) = \sum_{(i,j) \in P_\Delta} (m_{ij} - m'_{ij}) e^{\theta^\top z_{ij}}$ is a real-analytic function of θ that is not identically zero, and its zero set has Lebesgue measure zero in $\mathbb{R}^3$. Consequently, any two graphs with distinct count vectors are distinguished by $LO(\cdot; \theta)$ for almost every $\theta \in \mathbb{R}^3$, while only graphs with identical count vectors (Proposition 6) are indistinguishable for all θ .*

Proof. The map $(i, j) \mapsto z_{ij}$ is injective on P_Δ , since $\ln(ij)$ and $\ln(i + j)$ recover the product and the sum, which determine the unordered pair. Choose $v \in \mathbb{R}^3$ such that the inner products $v^\top z_{ij}$ are pairwise distinct, which excludes only finitely many hyperplanes of v . Along the ray $\theta = tv$, $D(tv) = \sum_k c_k e^{t\lambda_k}$ with distinct exponents λ_k and coefficients $c_k = m_{i_k j_k} - m'_{i_k j_k}$ that are not all zero, and such an exponential sum vanishes identically only if all $c_k = 0$. Hence $D \not\equiv 0$ on the ray, and so on $\mathbb{R}^3$. Since D is a finite sum of exponentials of linear functionals, it is real-analytic on $\mathbb{R}^3$, and the zero set of a nonzero real-analytic function has Lebesgue measure zero. ■

The proof uses only the injectivity of $(i, j) \mapsto (\ln(ij), \ln(i + j))$ and therefore applies to the parent $M_{\alpha, \beta}$ on the (α, β) -plane, so Proposition 7 is inherited from the parent rather than conferred by γ . Since $q_{ij} = \sqrt{1 - 4ij/(i + j)^2}$ depends only on the pair, γ adds no graph information beyond the edge-degree-pair counts, of which every vertex-degree-based index is a functional [30]. What it adds is the functional direction of Lemma 1, that is, a different weighting of the same information. Whether such a reweighting still helps prediction is the question of Section 6.

Table 1. The two degree-pair collision pairs among the 18 octane isomers. Each pair shares its complete edge-degree-pair count vector, shown as multiplicities of the unordered pairs, and is therefore indistinguishable by every vertex-degree-based index for all parameter values. The two members differ in the measured properties.

Isomer	T_B	ΔH_f	ΔH_{vap}	S	ω
<i>shared count vector</i> 2(1, 2), 1(1, 3), 2(2, 2), 2(2, 3):					
3-methylheptane	118.9	-50.82	9.52	111.26	0.371
4-methylheptane	117.7	-50.69	9.48	109.32	0.372
<i>shared count vector</i> 2(1, 2), 2(1, 3), 2(2, 3), 1(3, 3):					
3,4-dimethylhexane	117.7	-50.91	9.32	106.59	0.340
3-ethyl-2-methylpentane	115.6	-50.48	9.21	106.06	0.330

4.2 Degeneracy across tree orders

Exponential indices of this type were introduced by Rada [29] for their discriminating power on trees.

Degeneracy [21] measures the failure to discriminate non-isomorphic graphs, and we quantify it with the distinct-value convention $100(1 - k/N)$, where k is the number of distinct values among $N = 106$ trees (Supplementary Section S5). On the 106 trees of order 10 the pure- γ points $LO(0, 0, 1)$ and $LO(0, 0, 2)$ attain 25.5% (79 distinct values), tied with R , GA and AG . The 106 trees realise only 79 distinct edge-degree-pair count vectors, so 25.5% is the floor for *any* degree-pair-based index, while M_1 (83.0%) and M_2 (67.0%) are the most degenerate (Supplementary Figure S2). For orders 7 through 12, on all trees and on

those with $\Delta \leq 4$ (Supplementary Table S4), the pure- γ points and R attain the floor at every order. The index χ leaves it from order 10, with 77 values against 79 profiles. Theorem 2 makes this exact. The pure- γ points separate precisely the q -histograms, which coincide with the degree-pair profiles up to order 12 (order 15 for $\Delta \leq 4$) and not beyond. At order 13 the profiles $\{(1, 2)^1, (1, 3)^2, (1, 6)^5, (2, 3)^2, (2, 6)^1, (3, 3)^1\}$ and $\{(1, 3)^3, (1, 6)^5, (2, 2)^1, (2, 3)^2, (3, 6)^1\}$ (exponents are edge multiplicities) share the q -histogram $\{0^1, \frac{1}{5}^2, \frac{1}{3}^1, \frac{1}{2}^3, \frac{5}{7}^5\}$. Table 2 collects the first failures, and the index R still attains the floor at order 13 (570 values) and leaves it at order 14, where the pure- γ points give 1090 values.

Table 2. First tree order at which discrimination is lost. Values are counted as distinct at 11 significant digits. “Below floor” is the first order with fewer distinct values than distinct edge-degree-pair profiles, the floor for every vertex-degree-based index, and the entry gives (values/profiles) at that order. “Merges” is the first order at which an index on Π gives equal values to two trees with different q -histograms. This does not apply to R , which is not on Π . $LO(0, 0, \gamma)$ denotes the pure- γ points $\gamma = 1, 2$. Orders below 7 were not examined. A dash entry means not determined.

Index	all trees		trees with $\Delta \leq 4$	
	below floor	merges	below floor	merges
$LO(0, 0, \gamma)$	13 (566/570)	never ^a	16 (809/810)	never ^a
R	14 (1095/1100)	n/a ^b	> 16 ^c	n/a ^b
GA	13 (566/570) ^d	17 ^e	16 (809/810) ^d	— ^e

^a All q -histograms are separated for every nonzero algebraic γ (Theorem 2(ii)). ^b R is not on Π . ^c R attains the floor for trees with $\Delta \leq 4$ at every order up to 16, the largest examined. ^d GA lies on Π and merges no q -histograms below order 17, so there it ties the pure- γ points. ^e The witness (Section 3.1) has maximum degree 5.

Off the ratio plane the imbalance factor can restore discrimination lost at particular classical exponents. A point $(0, \beta, 0)$ weights an edge by its degree sum only, so pairs with equal sum receive equal weight. With $\gamma \neq 0$ their imbalances separate them. On all trees of order 16 (4069 profiles), $LO(0, \frac{1}{2}, 0)$ and $LO(0, \frac{1}{3}, 0)$ take only 3475 and 3476 distinct values, while $LO(0, \frac{1}{2}, 1)$ and $LO(0, \frac{1}{3}, 1)$ separate all 4069. $LO(\frac{1}{2}, 0, \cdot)$ rises from 3940 to 4067 and the Randić point $LO(-\frac{1}{2}, 0, \cdot)$ from 4007 to 4068 as γ goes from 0 to 1. Points on Π such as GA are unaffected (3996 in both cases),

as Theorem 2 requires, and generic parent points such as $LO(0.3, 0.2, 0)$ already separate all 4069 profiles (Proposition 7). The effect is established by enumeration for orders 12–16 but is not proved. It gives no predictive gain over a tuned parent in Section 6. At fixed fractional exponents the sign of the change depends on the property (Supplementary Section S5).

4.3 Structure sensitivity

The structure-sensitivity criterion of Furtula, Gutman and Dehmer [11], with a fingerprint-based variant due to Rakić and Furtula [31], was implemented with the published protocol on the 75 decane isomer trees. For a tree, the similar structures are those obtained by relocating a single edge within the molecular class. SS and Abr are the class means of the mean and the maximum relative change of the index over these neighbours. The implementation reproduces the published decane-class values of Barman and Das [5], with eleven of fourteen SS/Abr values agreeing exactly at the four displayed decimals. Each of the other three agrees within one unit in the fourth. The protocol and this control are detailed in Supplementary Section S6. Under the published desideratum (SS large, Abr small), $LO(0, 0, 2)$ has a structure sensitivity indistinguishable at three decimals from that of the hyper-Zagreb index (0.1257 against 0.1256). Its abruptness is markedly smaller (0.274 against 0.313). The sensitivity-to-abruptness ratios of the pure- γ points (0.4579 and 0.4565) are comparable to those of $H/2$, GA and R (0.448–0.451, Table 3). In view of the redundancy noted in Section 6.1, these third-decimal differences do not rank the indices. SS is also not a fixed property of the family. Along the pure- γ line it grows with γ (0.031, 0.062, 0.126, 0.319 at $\gamma = 0.5, 1, 2, 5$), while the ratio peaks near $\gamma = 2$ and then falls (0.427 at $\gamma = 5$). The pure- γ points reduce to first order in γ to the parameter-free aggregate $IRLA$, since $LO(0, 0, \gamma) = m + \frac{\gamma}{2} IRLA + O(\gamma^2)$, and this aggregate has a higher ratio ($SS = 0.2343$, $Abr = 0.5066$, ratio 0.4625). Hence the favourable sensitivity-abruptness balance is a property of linear imbalance weighting rather than of the exponential form. The equality of SS with hyper-Zagreb at $\gamma = 2$ reflects the chosen value of γ .

Table 3. Structure sensitivity SS , abruptness Abr , and their ratio on the 75 decane isomers under the published similar-structure protocol of [11] (single-edge relocation within the molecular class).

Index	SS	Abr	SS/Abr
M_1	0.0566	0.1336	0.4238
M_2	0.0922	0.2443	0.3776
HM	0.1256	0.3126	0.4017
mM_2	0.0516	0.1177	0.4384
R	0.0264	0.0588	0.4481
χ	0.0264	0.0591	0.4463
$H/2$	0.0516	0.1143	0.4514
ISI	0.0226	0.0597	0.3795
GA	0.0241	0.0535	0.4507
AG	0.0256	0.0579	0.4415
$LO(0, 0, 1)$	0.0624	0.1367	0.4565
$LO(0, 0, 2)$	0.1257	0.2745	0.4579

5 Closed forms, bounds and extremal trees

This section gives the closed-form values of the family on standard graphs, the extremal trees of the pure- γ points, and termwise bounds. The proofs are short and are included for completeness.

5.1 Closed-form values on standard graphs

The classical graph families used are the complete graph K_n , the cycle C_n , the path P_n , the star S_n and the complete bipartite graph $K_{p,q}$. Beyond these, the hypercube Q_k has 2^k vertices and $k \cdot 2^{k-1}$ edges and is k -regular. The wheel W_n is an $(n-1)$ -cycle plus a universal vertex, and the friendship graph F_p consists of p triangles sharing a common vertex. We also treat r -regular graphs.

Proposition 8. *Let $\alpha, \beta, \gamma \in \mathbb{R}$. Then:*

1. $LO(K_n; \alpha, \beta, \gamma) = \frac{n(n-1)}{2} 2^\beta (n-1)^{2\alpha+\beta}$ for $n \geq 2$.
2. $LO(C_n; \alpha, \beta, \gamma) = n \cdot 4^{\alpha+\beta}$ for $n \geq 3$.

3. $LO(P_n; \alpha, \beta, \gamma) = \begin{cases} 2^\beta, & n = 2, \\ 2 \cdot 2^\alpha 3^\beta e^{\gamma/3}, & n = 3, \\ 2 \cdot 2^\alpha 3^\beta e^{\gamma/3} + (n-3) \cdot 4^{\alpha+\beta}, & n \geq 4. \end{cases}$
4. $LO(S_n; \alpha, \beta, \gamma) = (n-1)(n-1)^\alpha n^\beta \exp\left(\gamma \frac{n-2}{n}\right)$ for $n \geq 2$.
5. $LO(K_{p,q}; \alpha, \beta, \gamma) = pq(pq)^\alpha (p+q)^\beta \exp\left(\gamma \frac{|p-q|}{p+q}\right)$.
6. $LO(Q_k; \alpha, \beta, \gamma) = k \cdot 2^{k-1} \cdot 2^\beta k^{2\alpha+\beta}$ for $k \geq 1$.
7. For $n \geq 5$, W_n has $n-1$ spoke edges with degree pair $(3, n-1)$ and $n-1$ rim edges with degree pair $(3, 3)$, so

$$LO(W_n; \alpha, \beta, \gamma) = (n-1) \left[3^\alpha (n-1)^\alpha (n+2)^\beta \exp\left(\gamma \frac{n-4}{n+2}\right) + 9^\alpha 6^\beta \right].$$
8. For $p \geq 1$, F_p has $2p$ spoke edges $(2, 2p)$ and p outer edges $(2, 2)$, so

$$LO(F_p; \alpha, \beta, \gamma) = p \cdot 4^{\alpha+\beta} + 2p(4p)^\alpha (2p+2)^\beta \exp\left(\gamma \frac{p-1}{p+1}\right).$$
9. For an r -regular graph G with m edges, $LO(G; \alpha, \beta, \gamma) = m \cdot 2^\beta r^{2\alpha+\beta}$.

Proof. Each case follows by enumerating the edge degree pairs and applying Definition 1. In K_n , C_n , Q_k and r -regular graphs every edge has $d_u = d_v = r$, so $\text{imb} = 0$ and $(d_u d_v)^\alpha (d_u + d_v)^\beta = 2^\beta r^{2\alpha+\beta}$, which gives items (1), (2), (6) and (9). All $n-1$ edges of S_n have degree pair $(1, n-1)$ with $\text{imb} = (n-2)/n$ (item (4)), and all pq edges of $K_{p,q}$ have pair (p, q) (item (5)). For $n \geq 4$, P_n has two pendant edges $(1, 2)$ and $n-3$ interior edges $(2, 2)$, and the cases $n = 2, 3$ are immediate (item (3)). Items (7) and (8) follow by summing over the two edge-degree-pair classes of W_n and F_p . ■

5.2 Extremal trees for the pure-imbalance points

The pure- γ points are ordered on trees by the same two graphs as most classical indices, and the proof is a short consequence of the edge-by-edge bound $0 \leq q_{uv} \leq (n-2)/n$ on a tree of order n .

Proposition 9 (Extremal trees). *Let $n \geq 4$ and $\gamma > 0$. Among all trees of order n , the path P_n is the unique minimiser and the star S_n the unique maximiser of $LO(\cdot; 0, 0, \gamma)$, with*

$$(n - 3) + 2e^{\gamma/3} \leq LO(T; 0, 0, \gamma) \leq (n - 1)e^{\gamma(n-2)/n}.$$

For $\gamma < 0$ the roles of P_n and S_n are exchanged.

Proof. Let $\gamma > 0$. Every edge uv of a tree of order n has $q_{uv} \leq (n - 2)/n$, with equality if and only if $\{d_u, d_v\} = \{1, n - 1\}$, so $LO(T; 0, 0, \gamma) \leq (n - 1)e^{\gamma(n-2)/n}$ with equality if and only if every edge is a $(1, n - 1)$ -edge, that is, $T = S_n$. For the lower bound, a tree with $k \geq 2$ leaves has at least k edges incident to a leaf, and each such edge has $q \geq \frac{1}{3}$ because the other end has degree at least 2 and $(d - 1)/(d + 1)$ is increasing in d . Hence $LO(T; 0, 0, \gamma) \geq (n - 1 - k) + ke^{\gamma/3} \geq (n - 3) + 2e^{\gamma/3}$, since $e^{\gamma/3} > 1$. Equality in the second inequality forces $k = 2$, so $T = P_n$, and P_n attains the bound because its two pendant edges are $(1, 2)$ -edges and its other edges are $(2, 2)$ -edges. For $\gamma < 0$ every inequality reverses. ■

The proof uses only the bounds on q and does not extend verbatim to general (α, β) , where the ordering of trees depends on the scale exponent $2\alpha + \beta$ as well as on γ . Among chemical trees ($\Delta \leq 4$) the termwise bound gives $LO(T; 0, 0, \gamma) \leq (n - 1)e^{3\gamma/5}$ for $\gamma > 0$, with equality only for $n = 5$. The path remains the unique minimiser for $\gamma > 0$ among chemical trees by Proposition 9, while the maximal chemical trees for $n > 5$ are left open. The orders 4 to 12 were also checked by enumeration in the reproducibility package.

5.3 Termwise bounds

We give a bracket bound via the edge-contribution function ψ , bounds in (m, Δ, δ) for non-negative parameters, a Cauchy–Schwarz interpolation bound and a uniform ratio bound. The $\gamma \neq 0$ bounds rest on bounding $e^{\gamma q_{uv}}$ edge by edge (for $\gamma \geq 0$) between 1 and $e^{\gamma(\Delta-1)/(\Delta+1)}$. Throughout, G is a simple connected graph with $\delta \leq d_v \leq \Delta$ for every $v \in V(G)$. Several of the inequalities have been machine-checked (Supplementary Section S7).

Remark (Attribution at $\gamma = 0$). At $\gamma = 0$ the family coincides with $M_{\alpha,\beta}$, and inequalities of all the types developed here are already known for it. These include (Δ, δ, m) -bounds with regular/biregular equality, two-sided comparisons against M_1 , M_2 , R , χ , H , GA , AG and ISI [24], Jensen- and Hölder-type relations [13] and further inequalities [28]. The $\gamma = 0$ specialisations below are therefore not claimed as new. What this section adds is their extension to $\gamma \neq 0$, where the imbalance factor $e^{\gamma q_{uv}}$ enters every edge weight, and the tracking of how γ deforms the equality cases. Comparisons at $\gamma = 0$ with the Randić, sum-connectivity and Sombor indices are not repeated.

The summand in Eq. (1) depends only on the unordered edge degree pair (d_u, d_v) . We isolate this dependence as follows:

Definition 2. For real parameters α, β, γ and positive reals x, y ,

$$\psi(x, y; \alpha, \beta, \gamma) = (xy)^\alpha (x + y)^\beta \exp\left(\gamma \frac{|x - y|}{x + y}\right). \quad (3)$$

With this notation $LO(G; \alpha, \beta, \gamma) = \sum_{uv \in E(G)} \psi(d_u, d_v; \alpha, \beta, \gamma)$.

Proposition 10. *Let G be a simple connected graph of size $m \geq 1$ with degrees in $[\delta, \Delta]$. Then*

$$m \cdot \psi_{\min}(\delta, \Delta; \alpha, \beta, \gamma) \leq LO(G; \alpha, \beta, \gamma) \leq m \cdot \psi_{\max}(\delta, \Delta; \alpha, \beta, \gamma), \quad (4)$$

where $\psi_{\min} = \min\{\psi(i, j; \alpha, \beta, \gamma) : \delta \leq i \leq j \leq \Delta\}$ and $\psi_{\max} = \max\{\psi(i, j; \alpha, \beta, \gamma) : \delta \leq i \leq j \leq \Delta\}$. The upper inequality is an equality if and only if every edge of G has a degree pair attaining $\psi_{\max}$. The lower inequality is an equality if and only if every edge has a degree pair attaining $\psi_{\min}$.

Proof. Every edge uv has its degree pair in the admissible set $\mathcal{P}(\delta, \Delta) := \{(i, j) \in \mathbb{Z}^2 : \delta \leq i \leq j \leq \Delta\}$, hence $\psi_{\min} \leq \psi(d_u, d_v; \alpha, \beta, \gamma) \leq \psi_{\max}$ on every edge, and summing over the m edges and using $\sum_{uv} \psi(d_u, d_v; \alpha, \beta, \gamma) = LO(G; \alpha, \beta, \gamma)$ gives (4). Each side of (4) is the sum of m per-edge inequalities, and is therefore an equality if and only if every per-edge inequality is. That is, equality holds if and only if every

edge has a degree pair attaining $\psi_{\max}$ (upper side), respectively $\psi_{\min}$ (lower side), over $\mathcal{P}(\delta, \Delta)$. $\blacksquare$

Corollary. *Let $\alpha_1 + \alpha_2 = 2\alpha$, $\beta_1 + \beta_2 = 2\beta$, and $\gamma_1 + \gamma_2 = 2\gamma$. Then for every simple connected graph G without isolated vertices,*

$$LO(G; \alpha, \beta, \gamma) \leq \sqrt{LO(G; \alpha_1, \beta_1, \gamma_1) \cdot LO(G; \alpha_2, \beta_2, \gamma_2)}$$

Equality holds if and only if the ratio

$$\psi(d_u, d_v; \alpha_1, \beta_1, \gamma_1) / \psi(d_u, d_v; \alpha_2, \beta_2, \gamma_2)$$

is the same for every edge of G .

Proof. For each edge $e = uv$ write $\psi_e^{(k)} := \psi(d_u, d_v; \alpha_k, \beta_k, \gamma_k) > 0$, $k \in \{1, 2\}$, and by additivity of the exponents and the hypotheses on $(\alpha_k, \beta_k, \gamma_k)$, $\psi_e^{(1)} \psi_e^{(2)} = \psi(d_u, d_v; \alpha, \beta, \gamma)^2$ on every edge. The Cauchy–Schwarz inequality for $a_e = (\psi_e^{(1)})^{1/2}$ and $b_e = (\psi_e^{(2)})^{1/2}$ gives $\sum_e a_e b_e \leq (\sum_e a_e^2)^{1/2} (\sum_e b_e^2)^{1/2}$, which is the claim because $\sum_e a_e b_e = LO(G; \alpha, \beta, \gamma)$, $\sum_e a_e^2 = LO(G; \alpha_1, \beta_1, \gamma_1)$ and $\sum_e b_e^2 = LO(G; \alpha_2, \beta_2, \gamma_2)$. Equality holds if and only if $b_e = \lambda a_e$ for a constant $\lambda \geq 0$ and all e , i.e. if and only if $\psi_e^{(2)} / \psi_e^{(1)} = \lambda^2$ is the same on every edge. (The inequality itself is midpoint log-convexity of $\theta \mapsto LO(G; \theta)$, Theorem 1.) $\blacksquare$

Proposition 11. *Let G be a connected graph of size m and maximum degree Δ . For $\alpha, \beta \geq 0$ and $\gamma \geq 0$,*

$$LO(G; \alpha, \beta, \gamma) \leq m \cdot \Delta^{2\alpha} \cdot (2\Delta)^\beta \cdot \exp\left(\gamma \frac{\Delta - 1}{\Delta + 1}\right). \quad (5)$$

Equality holds if and only if $\alpha = \beta = \gamma = 0$, or $\gamma = 0$, $\alpha + \beta > 0$ and G is Δ -regular, or $\gamma > 0$, $\alpha = \beta = 0$ and $G \cong K_{1, \Delta}$ with $\Delta \geq 2$; for $\gamma > 0$ and $\alpha + \beta > 0$ the inequality is strict.

Proof. On every edge uv we have $1 \leq d_u, d_v \leq \Delta$, hence $d_u d_v \leq \Delta^2$ and $d_u + d_v \leq 2\Delta$, and the normalised imbalance $\text{imb}(x, y) = |x - y| / (x + y)$ is maximised over $\{(i, j) : 1 \leq i \leq j \leq \Delta\}$ at $(1, \Delta)$, so $|d_u - d_v| / (d_u + d_v) \leq$

$(\Delta-1)/(\Delta+1)$. Since $\alpha, \beta, \gamma \geq 0$, the maps $t \mapsto t^\alpha$, $t \mapsto t^\beta$ and $t \mapsto e^{\gamma t}$ are non-decreasing, so $\psi(d_u, d_v; \alpha, \beta, \gamma) \leq \Delta^{2\alpha}(2\Delta)^\beta \exp(\gamma(\Delta-1)/(\Delta+1))$, and summing over the m edges yields (5). For $\gamma > 0$ and $\Delta \geq 2$ the bound is attained if and only if $\alpha = \beta = 0$ and every edge has degree pair $(1, \Delta)$ (for connected G , $G \cong K_{1,\Delta}$). If $\alpha > 0$ or $\beta > 0$ the three factor-wise maxima are not simultaneously attainable and the inequality is strict. ■

Proposition 12. *Let G be a connected graph of size m and minimum degree δ . For $\alpha, \beta \geq 0$ and $\gamma \geq 0$,*

$$LO(G; \alpha, \beta, \gamma) \geq m \cdot \delta^{2\alpha} \cdot (2\delta)^\beta, \quad (6)$$

with equality if and only if G is δ -regular, except in the degenerate case $(\alpha, \beta, \gamma) = (0, 0, 0)$, where $LO(G; \alpha, \beta, \gamma) \equiv m$ and the bound holds with equality for every graph.

Proof. On every edge $d_u, d_v \geq \delta$, so for $\alpha, \beta \geq 0$, $(d_u d_v)^\alpha \geq \delta^{2\alpha}$ and $(d_u + d_v)^\beta \geq (2\delta)^\beta$. For $\gamma \geq 0$ the exponential factor is at least 1 because the imbalance is non-negative, and multiplying and summing over the m edges yields (6). If $(\alpha, \beta, \gamma) = (0, 0, 0)$ then $LO(G; 0, 0, 0) = m$ equals the bound for every graph, while otherwise equality requires equality in every active per-edge factor, which for $\alpha > 0$ or $\beta > 0$ forces $d_u = d_v = \delta$ on every edge. If $\alpha = \beta = 0$ and $\gamma > 0$ it forces $d_u = d_v$ on every edge, so the connected graph G is regular, hence δ -regular. Conversely, on a δ -regular graph every edge has degree pair (δ, δ) and equality holds. ■

The next proposition gives a single uniform template for bounding $LO(G; \alpha, \beta, \gamma)$ both above and below. The bound is in terms of any classical bond-incident-degree index whose edge contribution is strictly positive on the admissible degree pairs of the graph.

Proposition 13. *Let G be a simple graph with maximum degree at most Δ . Let $I_h(G) = \sum_{uv \in E(G)} h(d_u, d_v)$ be a BID index with edge contribution $h(i, j)$ strictly positive on every degree pair (i, j) that appears in G . Define*

$$c_{\min}^h = \min_{(i,j)} \frac{\psi(i, j; \alpha, \beta, \gamma)}{h(i, j)}, \quad c_{\max}^h = \max_{(i,j)} \frac{\psi(i, j; \alpha, \beta, \gamma)}{h(i, j)},$$

where the extrema range over the degree pairs actually appearing in G . Then

$$c_{\min}^h \cdot I_h(G) \leq LO(G; \alpha, \beta, \gamma) \leq c_{\max}^h \cdot I_h(G), \quad (7)$$

with equality on either side if and only if every edge of G attains the corresponding extremum of ψ/h .

Proof. On every edge uv we have $h(d_u, d_v) > 0$ and $c_{\min}^h \leq \psi(d_u, d_v; \alpha, \beta, \gamma)/h(d_u, d_v) \leq c_{\max}^h$. Multiplying by $h(d_u, d_v)$ and summing over $E(G)$, with $\sum_{uv} h(d_u, d_v) = I_h(G)$ and $\sum_{uv} \psi = LO(G; \alpha, \beta, \gamma)$, gives (7). Each side is an equality if and only if every edge attains the corresponding extremum of ψ/h over the appearing pairs. ■

Proposition 13 is a *sharp graph-specific* comparison, whose constants depend on the degree pairs that actually appear in G , and a uniform version follows immediately by enlarging the range of the extrema.

Corollary (Uniform Δ -bounded version). *Assume the hypotheses of Proposition 13, but with h strictly positive on all of $P_\Delta = \{(i, j) : 1 \leq i \leq j \leq \Delta\}$ and the extrema taken over P_Δ . Then inequality (7) holds simultaneously for every graph of maximum degree at most Δ , with constants independent of the individual graph. Equality on either side holds if and only if every edge of G realises a degree pair attaining the corresponding extremum over P_Δ . The uniform constants are in general strictly worse than the graph-specific ones, since the extrema are taken over a superset.*

6 QSPR diagnostics

We benchmark the Loyola family against the classical indices it reduces to on the 18 octane isomers, following the chemometric benchmark of the diminished Sombor study [25], and use the 35 nonane isomers as a replication set with models refitted on it. The comparison set consists of the exact reductions of Section 1, $M_1, M_2, HM, {}^mM_2, R, \chi, H/2, ISI, GA, AG$, and the two fixed pure- γ benchmark points $LO(0, 0, 1)$ and $LO(0, 0, 2)$. Since the edge count $m = LO(0, 0, 0)$ is constant on the octanes, it is

omitted. No parameter in the correlation comparison is tuned against the property data, and an exploratory local-tuning check (Supplementary Section S2) did not exceed the best fixed benchmark except by a 0.001 in-sample ΔH_{vap} margin. All comparisons are descriptive. On sets this small they indicate how imbalance information changes the fit of the parent family, and they do not establish chemical predictive utility.

In Tables 5–7 medians are taken over candidate-stream seeds, seed counts describe the random search on fixed data rather than replications, and values are rounded once for display, so a tabulated difference may differ in the last place from the difference of two displayed entries. The narrow- and wide-box streams use the same uniform variates scaled by the half-width, so comparisons across boxes are paired.

6.1 Correlations on the octane isomers

Table 4 reports the Pearson correlation r between each index and the boiling point T_B , the enthalpy of formation ΔH_f , the enthalpy of vaporisation ΔH_{vap} , the entropy S and the acentric factor ω of the 18 octane isomers. The property values are the standard octane benchmark compilation, audited value by value against the NIST WebBook [23] and the 1947 API Research Project 44 tables [34]; Supplementary Table S5 lists them and Supplementary Section S1 records the status of each of the 90 values. Three items remain open. The 298 K vaporisation enthalpy of 2,2,3,3-tetramethylbutane, a solid at 298 K, has an unresolved physical reference state, six gas-phase entropies have no primary source, and the acentric factors are compiled values. Their effect on Table 4 is confined to the third decimal and to the 0.001 margin between HM and M_1 on S (Supplementary Section S8), and none of them enters the pre-registered tests of Sections 6.3 and 6.4.

The best correlation per property is split. ${}^mM_2 = LO(-1, 0, 0)$ is strongest on T_B and ΔH_f , $HM = LO(0, 2, 0)$ on S and $M_2 = LO(1, 0, 0)$ on ω , while the fixed pure- γ point $LO(0, 0, 1)$ is strongest on ΔH_{vap} ($|r| = 0.984$, with $LO(0, 0, 2)$ second at 0.981). At the exact-reduction triples the Loyola correlations coincide with the classical indices, as they must. Recomputing each bolded correlation with single isomers deleted

Table 4. Pearson correlation r between the Loyola reductions, the fixed pure- γ benchmark points and the five physicochemical properties of the 18 octane isomers. Bold flags the largest $|r|$ per column, and this is descriptive. Margins to the runner-up are 0.001–0.032, and their paired bootstrap intervals include zero (see the text).

Index	T_B	ΔH_f	ΔH_{vap}	S	ω
$M_1 = LO(0, 1, 0)$	−0.718	−0.762	−0.935	−0.970	−0.970
$M_2 = LO(1, 0, 0)$	−0.497	−0.542	−0.811	−0.934	−0.988
$HM = LO(0, 2, 0)$	−0.654	−0.710	−0.903	−0.971	−0.982
${}^mM_2 = LO(−1, 0, 0)$	+0.855	+0.891	+0.925	+0.863	+0.794
$R = LO(−1/2, 0, 0)$	+0.819	+0.850	+0.958	+0.932	+0.898
$\chi = LO(0, −1/2, 0)$	+0.800	+0.832	+0.961	+0.947	+0.924
$H/2 = LO(0, −1, 0)$	+0.821	+0.849	+0.961	+0.932	+0.900
$ISI = LO(1, −1, 0)$	−0.005	−0.000	−0.383	−0.593	−0.738
$GA = 2 LO(1/2, −1, 0)$	+0.822	+0.858	+0.965	+0.939	+0.906
$AG = \frac{1}{2} LO(−1/2, 1, 0)$	−0.816	−0.859	−0.957	−0.938	−0.900
$LO(0, 0, 1)$	−0.828	−0.832	−0.984	−0.928	−0.929
$LO(0, 0, 2)$	−0.829	−0.846	−0.981	−0.937	−0.925

shifts $|r|$ by at most 0.027 (Supplementary Section S1). The bolded entries are the largest of twelve highly correlated sample correlations on 18 molecules and are therefore optimistic estimates of the selected descriptor’s correlation on new molecules. They are descriptive rankings rather than tests. Fourteen of the descriptor pairs have $|r| \geq 0.99$ and the first two principal components carry 99.05% of the variance (Supplementary Section S1), and the pure- γ points are collinear with the classical reductions (maximum $|r|$ with a reduction equal to 0.982 for $LO(0, 0, 1)$ and 0.993 for $LO(0, 0, 2)$). Differences in the third decimal of $|r|$ between near-collinear descriptors are therefore not meaningful rankings.

Proposition 6 is realised on these data, where the 18 isomers realise only 16 distinct edge-degree-pair count vectors. The two collision pairs of Table 1 receive the same value from every vertex-degree-based index at every parameter choice, although their measured properties differ, by up to 1.2°C in T_B within the first pair and 2.1°C within the second. Every correlation in Table 4, and every tuned variant below, operates above this irreducible floor.

6.2 Effect of freeing γ and of the search box

We ask whether, at identical search budgets, freeing γ improves out-of-sample prediction beyond an optimised parent $M_{\alpha,\beta}$. The design is a fully nested leave-one-out (Supplementary Section S3). For each property and each outer fold, candidate parameter vectors are drawn uniformly per coordinate from a search box, triples (α, β, γ) for the Loyola model and pairs (α, β) with $\gamma = 0$ for the parent, with identical budget. Candidates are scored on the training isomers only, by inner leave-one-out root-mean-square error of the one-descriptor linear regression, after which the best candidate is refit and the held-out isomer predicted. No quantity computed on the held-out isomer enters selection or fitting.

In the box $(-2, 2)$ used first, freeing γ appeared to help the vaporisation enthalpy. It improved the nested Q^2 for ΔH_{vap} in 100/100 candidate streams at budgets 200 and 500, with median $\Delta Q^2 = +0.069$ and $+0.068$ (Supplementary Table S3), while it usually degraded ΔH_f and gave mixed results for the other properties. With 2,2,3,3-tetramethylbutane excluded the count falls to 93/100 and 96/100. These counts describe optimiser variance on one fixed set of 18 molecules, and they hide a boundary effect. For ΔH_{vap} the parent chose $(\alpha, \beta) = (1.021, -1.993)$, on the boundary of the box, in 16 of the 18 folds.

The decisive design choice is the box, which the nonane pre-registration also fixed at half-width 2. By $\ln(ij) = 2\ln(i + j) + \ln(1 - q^2) - \ln 4$ (Eq. (2)) the parent reaches imbalance information through $(1 - q^2)^\alpha$ along the direction $\beta \approx -2\alpha$, which needs large exponents and is exactly the direction the box truncates. On a dense parent grid (steps 0.05, 0.1, 0.2 and 0.25) the nested Q^2 of the parent for octane ΔH_{vap} is 0.8873, 0.9156, 0.9525 and 0.9383 at half-widths 2, 4, 8 and 12. The selected point lies within one grid step of the boundary in 18, 18, 2 and 0 of the 18 folds. On the nonanes the values are 0.8532, 0.8779, 0.8997 and 0.8837, with 34, 32, 3 and 0 boundary folds of 34. The half-width-12 box, chosen after inspecting these two ΔH_{vap} curves as the smallest examined in which the parent’s selections are interior, is used below for every property. The matched comparison was therefore repeated with 2000 candidates per model, seeds 0–49, and boxes of half-width 2 and 12 for both models (Table 5); this

repetition was not pre-registered. In the table, Edge is the share of outer folds, pooled over the 50 seeds, in which the parent’s selection lies within 10% of the boundary of the box. The counts are the number of seeds in which LO beats GM (half-width 2) or LO_h beats GM (half-width 12). In the wide box the vaporisation-enthalpy gain shrinks to about zero. A hybrid control widens only (α, β) and keeps $\gamma \in (-2, 2)$; it is drawn after the parent from the same candidate stream, so the two are paired by seed rather than by candidate. It gives median $\Delta Q^2 = +0.004$ on the octanes (28 of 50 seeds) and -0.002 on the nonanes (24 of 50), while widening all three coordinates gives -0.010 and -0.016 . On the other properties no advantage reaches the seed-count criterion in either box. Nonane T_B has 30 and 31 wins of 50 in the two boxes, and octane S moves from 11 to 38, which illustrates optimiser variance rather than an effect. Most of the change comes from the parent improving in the wide box ($0.887 \rightarrow 0.942$ on the octanes), and the rest from the three-dimensional search being slightly less efficient there ($0.957 \rightarrow 0.946$ for the hybrid control). What remains ($+0.004$ in 28 of 50 seeds) is within optimiser variance. The fixed point $LO(0, 0, 1)$ and the parameter-free *IRLA* match the fixed parent point $M_{4.5,-9}$ (leave-one-out $Q^2 = 0.957$ and 0.948 against 0.949 , on the octanes), which lies near the parent’s wide-box selections.

Table 5. Sensitivity of the matched comparison to the search box (medians of Q^2 over seeds 0–49 at 2000 candidates per model).

Set, prop.	half-width 2				half-width 12				
	Edge	GM	LO	>	Edge	GM	LO	LO_h	>
oct. T_B	69%	0.665	0.730	36	2%	0.606	0.495	0.544	21
oct. ΔH_f	52%	0.821	0.769	11	0%	0.820	0.670	0.761	10
oct. ΔH_{vap}	97%	0.887	0.957	50	3%	0.942	0.934	0.946	28
oct. S	85%	0.904	0.891	11	4%	0.842	0.838	0.867	38
oct. ω	0%	0.979	0.974	10	0%	0.979	0.938	0.951	7
non. T_B	61%	0.576	0.601	30	0%	0.524	0.392	0.577	31
non. ΔH_{vap}	93%	0.855	0.895	49	9%	0.892	0.881	0.889	24

Under these protocols tuning γ yields no demonstrated predictive advantage over a parent evaluated over the wider box. Two caveats remain. Equal candidate budgets do not guarantee equally thorough searches in

two and three dimensions, and the hybrid control reduces this asymmetry without removing it, so the matched design costs the three-parameter model part of the effect sought. The parent’s own Q^2 also varies with the box by about as much as the narrow-box gain, as the grid values above show. We conclude that no gain has been demonstrated, which leaves open whether a gain exists.

6.3 Replication on the nonane isomers

Under a protocol committed to the reproducibility package (commit 964b127) before any nonane value was collected, the identical nested design was applied to the 35 constitutional nonane isomers. Boiling points were 35 values, namely 20 NIST averages, 5 single determinations and 10 most recent determinations selected, where NIST gives no average, as the most recent determination (a deviation recorded in the protocol). Standard vaporisation enthalpies were 34 values from the NIST WebBook [23]. Only 4 of them are calorimetric determinations (one more is a NIST average), and 29 come from two compilations, the 1961 tabulation of Labbauf et al. [22] and the 1971 Wilhoit–Zwolinski handbook [39], which NIST lists under Reid’s 1972 review. Agreement with them therefore measures consistency with compiled values rather than independent experimental validation, and, if the compiled values rest on structural increments, it favours any model linear in the degree-pair counts. The pre-registered verdict is that the γ effect *replicates* for ΔH_{vap} , with LO better than GM in 95/100 and 98/100 seeds and median $\Delta Q^2 = +0.048$ and $+0.053$, and is *mixed* for T_B (58/100 and 63/100). Within-fold baselines are in Table 6. The protocol, however, fixed the box of half-width 2, which binds the parent (Table 5), and in a box that does not, the nonane gain disappears (median $\Delta Q^2 = -0.002$ under the hybrid control). The pre-registered verdict is reported as obtained, with this deviation recorded. The seed-count criterion of that protocol counts optimiser variance on fixed data, and we now regard it as a design error of the protocol rather than as evidence. Because the nonane vaporisation enthalpies are compilation values, we decided after this result to treat the measured boiling points of Section 6.4 as the primary predictive evidence

of the paper and the nonane analysis as a secondary diagnostic. That reclassification was made after the result, and we record it here.

On both sets the untuned fixed point $LO(0, 0, 1)$ matches or beats the tuned Loyola model (0.957 and 0.903 against medians 0.953 and 0.900). It is closely related to *IRLA*, since $LO(G; 0, 0, \gamma) = m + \frac{\gamma}{2} IRLA(G) + O(\gamma^2)$ as $\gamma \rightarrow 0$. At $\gamma = 1$ higher moments of the imbalance remain and trees with equal *IRLA* can differ in $LO(0, 0, 1)$, yet the correlation between the two is 0.996 on both sets, and *IRLA* itself comes within 0.003 of the median Loyola model on the nonanes (0.897 against 0.900). What matters for this property is therefore a term linear in the imbalance. Ridge regression on the full degree-pair count vector, a reference for what the counts carry rather than a competing index, predicts the nonanes better than every index model ($Q^2 = 0.927$ for ΔH_{vap} , 0.682 for T_B). A post hoc molecule bootstrap conditional on the fitted predictions, which omits refitting and selection uncertainty, gives a 95% interval for LO minus GM on nonane ΔH_{vap} of $[-0.006, +0.113]$ at budget 200, which includes zero (Supplementary Section S4).

Table 6. Nested leave-one-out Q^2 on the octanes (oct., $n = 18$) and the nonane replication set (non., $n = 35$ for T_B , 34 for ΔH_{vap}). Mean: training-fold mean. Class.: best of the ten classical indices, selected inside each fold. *IRLA*: one-descriptor model on *IRLA*. Ridge: ridge regression on the edge-degree-pair counts. LO_1 : the fixed, untuned point $LO(0, 0, 1)$. GM, LO: median of Q^2 over seeds 0–99 at budget 200 in the box of half-width 2 (see Table 5 for wider boxes). Last column: seeds in which LO beat GM at budgets 200/500.

Set, property	Mean	Class.	<i>IRLA</i>	Ridge	LO_1	GM	LO	LO>GM
oct. T_B	−0.121	0.593	0.585	0.344	0.591	0.662	0.652	53/64
oct. ΔH_f	−0.121	0.746	0.593	0.654	0.639	0.831	0.689	14/24
oct. ΔH_{vap}	−0.121	0.884	0.948	0.954	0.957	0.888	0.953	100/100
oct. S	−0.121	0.905	0.733	0.721	0.785	0.904	0.889	34/31
oct. ω	−0.121	0.971	0.810	0.851	0.823	0.979	0.970	26/28
non. T_B	−0.060	0.502	0.400	0.682	0.409	0.577	0.586	58/63
non. ΔH_{vap}	−0.062	0.857	0.897	0.927	0.903	0.853	0.900	95/98

6.4 Measured boiling points

Measured vaporisation enthalpies are too scarce for a larger test. NIST lists only 25 acyclic alkane skeletons outside C_8 and C_9 , and one of the 75 decanes, with at least one calorimetric standard-state 298 K determination. Two further tests therefore used measured boiling points, under a protocol committed to the repository (commit [a5cf378](#)) before any model was fitted. The NIST pages had been archived two days earlier during a data-availability audit that counted available values only. Boiling points were taken from the NIST WebBook [23] by a fixed rule. The rule takes the NIST average, or else the median of non-compilation determinations if they agree within 3 K, and otherwise excludes the molecule. This gives all hexanes, heptanes and octanes and 34 of the 35 nonanes, since one nonane has two NIST entries with the same InChI whose boiling points differ by 6.4 K. It also gives 34 of the 75 decanes, because 31 of them have no NIST boiling point and 10 have determinations more than 3 K apart. For 14 of the 34 included decanes NIST lists a single non-compilation determination, so the agreement rule does not apply to them. Under this uniform rule the pooled octane boiling points can differ from those of Supplementary Table S5 by up to 0.5 °C, for example 2,4-dimethylhexane at 108.9 against 109.4 °C, a documented source variation. Seven pooled nonane boiling points differ from the replication-set values of Section 6.3 by 0.1–0.9 °C, as that protocol took the most recent determination where NIST gives no average. The included and excluded decanes do not differ detectably in branching or imbalance, measured by the numbers of tertiary and quaternary carbons and of methyl groups, the Wiener index, *IRLA*, *LO*(0,0,1) and *R*. Two-sided permutation tests gave $p \geq 0.10$ in every case, though with 34 and 41 molecules such tests can rule out only large differences. Test A evaluates the models on the decanes as a new cohort, refitting and reselecting inside each leave-one-out fold. Test B pools all 100 molecules and adds the carbon number n_C as a covariate, so that each descriptor is judged on branching beyond size. Because the octane and nonane boiling points had already been used in Sections 6.1 and 6.3, Test B is a pooled size-adjusted analysis and not a fresh external test; only the hexanes, heptanes and decanes are new. Both use the non-truncating

design of Section 6.2, with the parent searched over $(\alpha, \beta) \in (-12, 12)^2$ and the Loyola model over the same box with $\gamma \in (-2, 2)$, at 2000 candidates and seeds 0–49. The pre-registered criterion for a γ effect required at least 40 wins in 50 seeds and a molecule-bootstrap interval excluding zero (Table 7). The rule is one-sided, so a negative difference also returns “no evidence” and the rule never returns evidence that γ hurts.

Table 7. Measured boiling points (nested leave-one-out Q^2). Test A: decane cohort, refitted within folds. Test B: pooled C6–C10 with the carbon number n_C in every model (Class., *IRLA*, LO₁: n_C plus that descriptor). Base: training-fold mean (Test A) or n_C alone (Test B). Class.: best classical index chosen in each fold. LO₁: fixed $LO(0, 0, 1)$. Ridge: ridge regression on the degree-pair counts. GM, LO: medians over seeds 0–49 in the non-truncating boxes. ΔQ^2 : median of LO minus GM. Last column: seeds (of 50) in which LO beats GM.

Test	Base	Class.	<i>IRLA</i>	LO ₁	Ridge	GM	LO _h	ΔQ^2	>
decanes ($n = 34$)	-0.062	0.176	0.167	0.149	0.860	0.514	0.406	-0.072	15
C6–C10, with n_C ($n = 100$)	0.938	0.967	0.963	0.963	0.991	0.977	0.977	-0.000	25

Both pre-registered verdicts are negative. On the decane cohort the median $\Delta Q^2 = -0.072$, LO is ahead in 15 of 50 seeds, and the 95% interval is $[-0.156, +0.001]$. In the pooled size-adjusted set the values are -0.000 , 25 of 50 and $[-0.003, +0.002]$. In the original box of half-width 2, reported as the pre-registration requires, the two tuned models are level on the decanes (medians 0.469 and 0.465). The Loyola model is marginally ahead on the pooled set (0.980 against 0.976). The decane verdict therefore does not depend on the box. The lower decane Q^2 of the hybrid control (0.406 against 0.465 in the narrow box) is consistent with the cost of a three-dimensional search over a wide box; the one-sided design cannot separate that cost from a penalty of selecting γ . The pooled cohort was corrected after an external audit found one nonane’s second NIST entry skipped (a deviation recorded in the repository), and pooling grouped pages also moved one heptane value by 0.2°C . On the uncorrected cohort of 101 molecules the same test gave -0.000 , 24 of 50, $[-0.003, +0.002]$, so the correction changes no conclusion. The intervals are molecule bootstraps conditional on the fitted predictions, as in Section 6.3. Ridge regression

on the ten degree-pair counts reaches $Q^2 = 0.860$ on the decanes, where every single index stays between 0.15 and 0.51, and 0.991 on the pooled set. A permutation check on the decanes (200 random permutations of the boiling points) gives it a median Q^2 of -0.09 , so the gap is not an artefact of the cross-validation procedure itself, although a permutation test cannot exclude every form of leakage. Branching effects on the boiling point are thus spread over several degree-pair counts in a way that no one-dimensional summary of the family, with or without γ , recovers. These conclusions concern acyclic alkanes.

Data, scripts and verifiers that recompute every tabulated value are in the reproducibility package [26], where the nonane and boiling-point protocols were committed before the corresponding models were fitted. The termwise bounds, the convexity statement, the independence of the imbalance direction and the order-thirteen profile collision have also been machine-checked in Lean 4 with Mathlib (Supplementary Section S7).

7 Conclusion

We extended the Gutman–Milovanović family by an exponential factor in the normalised degree imbalance of each edge and asked what this third coordinate adds. On the mathematical side we showed that the imbalance direction is affinely independent of the two parent directions once degree pairs containing a vertex of degree three are admitted, while on an individual graph the identifiable directions were characterised separately. The logarithm of the index is globally convex. The family factorises into a size coordinate and a shape coordinate. On the resulting ratio plane the pure-imbalance points discriminate exactly the histogram of edge imbalances, and the same limit binds every index that depends on an edge only through its degree ratio. For trees of small order this histogram carries the full degree-pair information, so these points attain the discrimination limit of the whole degree-based class. From order thirteen onward (sixteen for chemical trees) they provably lose information, and we exhibited the colliding families. We also bounded the number of order exchanges between two chemical graphs along the imbalance axis and determined the extremal

trees of the pure-imbalance points. For each graph we characterised the parameter directions that leave the index unchanged up to scale.

On the chemometric side no predictive gain was demonstrated. An apparent gain from the imbalance parameter on vaporisation enthalpies replicated under a pre-registered protocol, but it disappeared once the parent family was searched over a box that did not truncate it. Pre-registered tests on measured boiling points of decanes and of a pooled set of C6–C10 alkanes showed no demonstrated benefit from the third coordinate. There a linear model on the degree-pair counts outperformed every single index. The untuned pure-imbalance point is highly correlated with the parameter-free irregularity index and matched the tuned models on vaporisation enthalpy. The compactness of the family therefore comes from linear imbalance weighting rather than from tuning. The third coordinate remains useful as an interpretable direction on the parent family along which the effect of degree imbalance can be analysed exactly. Three problems remain open. The first is sharp two-sided bounds for the family in terms of indices outside it, such as the atom-bond-connectivity and Sombor indices. The second is the extremal graphs for general parameter triples, beginning with the maximal chemical trees of the pure-imbalance points, and for unicyclic and bicyclic graphs. The third is the chemometric behaviour on other compound classes, such as benzene derivatives, and on class-specific endpoints.

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